THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** 50	$7^2, 8^2, 9^2$ are squares
2	**(c)** 64	8×8
3	**(d)** 21	the rule gives 21
4	**(a)** two end points	fixed length
5	**(b)** 90°	a right angle
6	**(a)** acute	compare with 90°, 180°, 360°
7	**(c)** 6174	reached by the Kaprekar routine
8	**(c)** 6	apply the rule repeatedly
9	**(d)** data	raw facts
10	**(a)** Label both axes and write a title	good practice
11	**(d)** highest common factor	definition
12	**(a)** 5	common factors of 10 and 35

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	1, 3, 6, 10, 15.	1+1
14	$2 \times 30^\circ = 60^\circ$.	1+1
15	Even; e.g. 3 + 5 = 8.	1+1
16	$4.5 \times 20 = 90$.	1+1
17	A number with exactly two factors, 1 and itself; 2, 3, 5.	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Add 3 each time (1); 16, 19, 22 (2).	1+2
19	Take 68 + 86 = 154; 154 + 451 = 605; 605 + 506 = 1111 (2); a palindrome (1).	2+1
20	65 mm (1.5); 25 mm (1.5).	1½+1½
21	2 × 3 × 3 × 5 (2); four prime factors (1).	2+1
22	It misleads the eye (1.5); bars must be equal width and spacing (1.5).	1½+1½
	OR: Mango (1.5); total 42 (1.5).	1½+1½
23	Digit sum divisible by 9 (1); 7+1+2+8 = 18, so divisible (2).	1+2
	OR: Any common factor of n and n+1 divides their difference 1 (1.5); e.g. 8 and 9 are co-prime (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Acute, right, obtuse, straight, reflex, complete with measures (3). (ii) Examples (2).	3+2
	OR: (i) Definitions and diagrams (3). (ii) Examples (2).	3+2
25	(i) Rules (3). (ii) Divisible by 2, 3, 4, 5, 6, 9, 10 (2).	3+2
	OR: (i) $2^2 \times 3^2 \times 5$ (2). (ii) 18 (1.5). (iii) 1, 2, 3, 4, 5, 6, 9, 10, 12, 15, 18, 20, 30, 36, 45, 60, 90, 180 (1.5).	2+1½+1½

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET B

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** pattern rule	describes sequence
2	**(d)** 225	$n^2 = 225$
3	**(c)** right angle	90°
4	**(b)** obtuse	between 90° and 180°
5	**(d)** perpendicular lines	90°
6	**(b)** 1089	famous trick
7	**(d)** 8531	descending digits
8	**(b)** calculating quickly in the mind using tricks	easy strategies
9	**(c)** four vertical lines crossed by a slanting line	counting aid
10	**(a)** asking each child	survey
11	**(a)** 0	rule
12	**(b)** 6	common factors of 12 and 18

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	A list of numbers or shapes in a particular order following a rule; e.g. 2, 4, 6, 8.	1+1
14	Acute, right, obtuse, straight, reflex and complete angle.	1+1
15	7,500 and 7,000.	1+1
16	Pictures represent data; the key tells what one picture stands for.	1+1
17	7, 14, 21, 28, 35.	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Dots equal to each number can be arranged in a triangle with 1, 2, 3 … dots in rows (1½); each new number adds one more row (1½).	1½+1½
19	Minute hand: $20 \times 6 = 120^\circ$ (1.5); hour hand: 10° (1.5).	1½+1½
20	3½, 5, 2 symbols (3).	1+2
21	84 = 2×2×3×7; 126 = 2×3×3×7 (2); HCF = 42 (1).	2+1
22	Write numbers 1 to 100 (1); cross out 1 and multiples of 2, 3, 5, 7 keeping the prime (1); remaining numbers are primes (1).	1+1+1
	OR: 12 = 2²×3; 18 = 2×3² (1.5); LCM = 2²×3² = 36 (1.5).	1½+1½
23	Multiples of 3: $3 \times 20 = 60$ (3).	1+2
	OR: The next shape has 16 dots (1); a 4 by 4 grid (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Shapes following a rule (1). (ii) 1, 4, 9 squares (2). (iii) 25 (2).	1+2+2
	OR: (i) 7 (1). (ii) 28 (1). (iii) Triangular numbers (1). (iv) 8 rows (2).	1+1+1+2
25	(i) Numbers that read the same both ways (1). (ii) 161, 242, 323, 404 (2). (iii) 247 + 742 = 989 (2).	1+2+2
	OR: (i) 8+16 = 24, 9+15 = 24 (1). (ii) Equal (1). (iii) 12, 13, 19, 20: 32 and 32 (1). (iv) Moving down adds 7 and right adds 1 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET C

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** 66	$1+2+\cdots+11=66$
2	**(b)** 3, 4, 5, 6, …	one more each time
3	**(a)** obtuse	compare with 90°, 180°, 360°
4	**(b)** dot	marks a position
5	**(b)** 99999	all 9s
6	**(c)** 8	count of moves
7	**(d)** 7000	estimation
8	**(a)** how many items one symbol stands for	scale
9	**(b)** 3	counts
10	**(b)** 6 symbols	$96 \div 16$
11	**(a)** neither prime nor composite	only one factor
12	**(c)** less than or equal to the number	property

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$7^3 = 343$.	1+1
14	Points lying on the same line, e.g. three dots drawn along a ruler's edge.	1+1
15	5, 16, 8, 4, 2, 1.	1+1
16	A table showing how many times each observation occurs.	1+1
17	20.	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Triangular numbers (1.5); next is 15 (1.5).	1½+1½
19	Triangle of three points (1); 3 lines (2).	1+2
20	e.g. 1221, 3443; each palindrome is its own reverse (1.5); sum is twice the number (1.5).	1½+1½
21	1, 2, 3, 4, 6, 9, 12, 18, 36 (2); 9 factors (1).	2+1
22	1, 8, 27, 64, … (1); each is a number multiplied by itself thrice (1); they can be shown as cubes of unit blocks (1).	1+1+1
	OR: Triangular number: 55 (1); pairs $(1+10), (2+9)\ldots$ make 5 pairs of 11 (2).	1+2
23	$4 \times 30 = 120^\circ$ (1.5); at 8 o'clock also 120° (1.5).	1½+1½
	OR: Place centre at the vertex (1); align one arm with the zero line (1); read the scale where the other arm crosses (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Definitions (2). (ii) Correct diagrams (2). (iii) Point 0; segment 2; ray 1; line 0 (1).	2+2+1
	OR: (i) 20° (1). (ii) Both acute (1). (iii) The ladder would be vertical (1). (iv) Protractor (2).	1+1+1+2
25	(i) 36, 60, 24 (2). (ii) 120 (1). (iii) Wednesday (1). (iv) 36 (1).	2+1+1+1
	OR: (i) Steps (2). (ii) 4, 7, 3 symbols (2). (iii) 15 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.